Synthetic Fibres and Plastics

- **Synthetic fibres (or man-made fibres):**

    - They are chains of small units joined together (each small unit is a chemical substance).
    - These small units combine to form a large single unit called a polymer.
    - Types of polymers:

1. **Addition polymers**: monomers combine together to form a giant molecule known as the polymer. No molecule is eliminated during formation of addition polymers. Examples of addition polymer are polythene, polyvinyl chloride etc.

2. **Condensation polymers**: several small units of monomers combine with each other, along with elimination of simple molecule like water to form polymer unit. Examples of condensation polymer are nylon-66, terylene etc.

- **Types of Synthetic fibres:**

- **Rayon (or artificial silk):**
    - It is obtained from chemical treatment of wood pulp
    - It is mixed with cotton and is used to make bed sheets or mixed with wool to make carpets.

- **Nylon**
    - It is strong, elastic, and light.
    - A nylon thread is stronger than a steel wire.
    - It is used for making clothes, parachutes and ropes for rock climbing.

- **Polyester**
    - Fabric made from polyester does not get wrinkled easily.
    - Common polyester includes terylene and PET
    - PET is used for making utensils, films, wires, bottles, etc. Terylene is used for making dress materials.

- **Acrylic**
    - It is relatively cheaper than wool.
    - Sweaters, shawls and blankets are made from acrylic.

- **Characteristics of synthetic fibres:** They dry up quickly, are durable, less expensive, readily available, and easy to maintain. However, fabric made of synthetic fibre melts on catching fire and sticks to the body of person wearing it. So, synthetic clothes should not be worn while working in kitchen or laboratory.

- **Plastics:**

- Are also polymer-like synthetic fibres.
- Arrangement of small units is linear or cross-linked.
- Can be recycled, reused, coloured, melted, rolled into sheets, or made into wires.

- **Thermoplastics:** These are the plastics that get deformed easily on heating and can be bent easily. Examples: polythene and PVC

- **Thermosetting plastics:** These are the plastics, which when moulded once, cannot be softened by heating. Examples: bakelite and melamine

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  - They are light, strong, and durable.
  - They are poor conductors of heat and electricity.

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- **Use of plastics:**
  - They are used to store various kinds of materials such as food items, chemicals etc.
  - It is widely used in various industries and for making a variety of household articles.
  - They are extensively used in health care industry for making syringes, threads for stitching wounds, doctor's gloves, and other medical instruments.
  - Fire resistant plastics are used as a coating on the suits of the firemen.

- **Biodegradable substances:** These are the materials that decompose through natural processes such as by the action of bacteria. Examples: paper, peels of vegetables, wood and fruits, etc.

- **Non-biodegradable substances:** These are the materials that are not easily decomposed by natural processes. Examples: plastic bags, metals, etc.
  - Plastics are not environment friendly as they cause environment pollution.
  - To minimize the environmental hazards, the **4 R** principle must be used.
  - **Reduce, Reuse, Recycle,** and **Recover.**